THE EXTEMPORE LANGUAGE: ON-THE-FLY COMPILATION FOR THE WHOLE AUDIOVISUAL STACK.

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ABSTRACT

Extempore is a systems programming language designed to support the programming of real-time systems. Extempore promotes human orchestration as a meta-model of real-time man-machine interaction in an increasingly distributed and environmentally aware computing context. Extempore is designed to support a style of programming dubbed cyberphysical programming [1].

Extempore is ideally suited for on-the-fly audiovisual software development as the language affords runtime modification of the entire audiovisual application stack. In practice this translates into a system capable of building, executing and modifying audiovisual code at both the control and signal layer in live performance. Extempore removes any divide between signal and control layers without sacrificing the performance required to build applications which remain 'close to the metal'.

Extempore is a general purpose environment and is not designed primarily as a signal processing language. This makes the environment flexible and suitable for a wide range of real-time systems programming roles. In particular Extempore is designed to support heterogeneous systems and architectures from mobile devices through to high performance clusters. This generality proves useful when building audiovisual systems designed for distributed systems and has been used extensively in QUT's new distributed 'cube' project where Extempore has formed the foundation for a 24 node real-time rendering system.

Extempore supports a wide range of audiovisual tasks including audio/visual/scientific DSP, numerical computation, algorithmic composition, 2D vector graphics, 3D model loading and animation, a 3D scene graph, gpu image processing, video processing, bullet physics integration etc..

Extempore is nearing a v1.0 release scheduled for September 2012 and is available for Linux, Windows 7, and OSX. Extempore is an open source project hosted at http://www.github.com/digego/extempore.

1. REFERENCES

[1] A. Sorensen and H. Gardner, “Programming with time: cyber-physical programming with impromptu,” in Proceedings of ACM OOPLSA, 2010, pp. 822–834.